

Demo Exercise Sheet 11

Mirror Descent, Projected Gradients, and FTRL

Operations Research

10th July

Roadmap for the Tutorial

1. Mirror descent

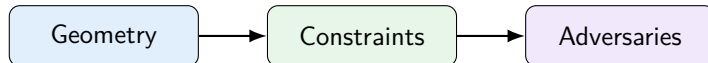
Bregman divergence, dual-space update, entropy geometry.

2. Projected gradients

Gradient step, projection step, unit-disk example.

3. Online optimization

FTL, BTL, FTRL, regret, and regularization.



Exercise CE11.1: Mirror Descent and Bregman Divergence

Mirror descent generalizes gradient descent by replacing the Euclidean distance with a Bregman divergence $D_h(x||y)$ generated by a strictly convex function $h(x)$. The primal update is:

$$x^{(k+1)} = \arg \min_x \{ \langle \eta \nabla f(x^{(k)}), x \rangle + D_h(x||x^{(k)}) \}$$

which leads to the dual update: $\nabla h(x^{(k+1)}) = \nabla h(x^{(k)}) - \eta \nabla f(x^{(k)})$.

- (a) Let $h(x) = \frac{1}{2} \|x\|_2^2$. Compute the Bregman divergence $D_h(x||y)$ and show that the Mirror Descent update reduces to standard Gradient Descent.
- (b) Let $h(x) = \sum_{i=1}^n (x_i \log x_i - x_i)$. Compute the gradient $\nabla h(x)$ and its inverse $(\nabla h)^{-1}(y)$.
- (c) Using the function h from (b), derive the coordinate-wise update rule for $x_i^{(k+1)}$. How does this relate to the Hedge (Exponential Weights) algorithm?

CE11.1: Main Idea

Why not always use Euclidean geometry?

Gradient descent measures movement with $\|x - y\|_2^2$. This is natural in Euclidean space, but often unnatural on structured domains, such as probability vectors.

Gradient descent

Use the squared Euclidean distance:

$$\frac{1}{2} \|x - y\|_2^2.$$

The update is additive.

Mirror descent

Use a geometry generated by a strictly convex function h :

$$D_h(x \| y).$$

The update is additive in the dual space.

CE11.1: Bregman Divergence

Definition

For a differentiable, strictly convex function h , the Bregman divergence from y to x is

$$D_h(x\|y) = h(x) - h(y) - \langle \nabla h(y), x - y \rangle .$$

Geometric interpretation

$D_h(x\|y)$ is the vertical gap between $h(x)$ and the first-order Taylor approximation of h around y :

$$h(y) + \langle \nabla h(y), x - y \rangle .$$

CE11.1: Mirror Descent Update

Primal form

Given $x^{(k)}$, choose

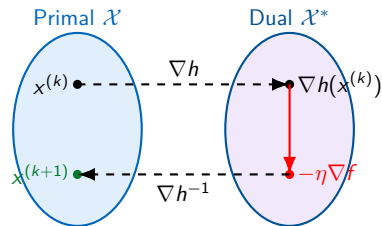
$$x^{(k+1)} = \arg \min_x \left\{ \left\langle \eta \nabla f(x^{(k)}), x \right\rangle + D_h(x \| x^{(k)}) \right\}$$

First-order condition

$$\eta \nabla f(x^{(k)}) + \nabla h(x^{(k+1)}) - \nabla h(x^{(k)}) = 0$$

Dual-space update

$$\nabla h(x^{(k+1)}) = \nabla h(x^{(k)}) - \eta \nabla f(x^{(k)})$$



CE11.1(a): Squared Euclidean Geometry

Task

Let

$$h(x) = \frac{1}{2} \|x\|_2^2.$$

Compute $D_h(x\|y)$ and the mirror descent update.

Gradient of the mirror map

$$\nabla h(x) = x.$$

So the dual space and the primal space are identical.

CE11.1(a): Bregman Divergence for $h(x) = \frac{1}{2} \|x\|_2^2$

$$\begin{aligned} D_h(x\|y) &= h(x) - h(y) - \langle \nabla h(y), x - y \rangle \\ &= \frac{1}{2} x^\top x - \frac{1}{2} y^\top y - y^\top (x - y) \\ &= \frac{1}{2} x^\top x - \frac{1}{2} y^\top y - y^\top x + y^\top y \\ &= \frac{1}{2} x^\top x - y^\top x + \frac{1}{2} y^\top y \\ &= \frac{1}{2} \|x - y\|_2^2. \end{aligned}$$

Conclusion

The Euclidean Bregman divergence is exactly the squared Euclidean distance.

CE11.1(a): Mirror Descent Becomes Gradient Descent

From the dual update,

$$\nabla h(x^{(k+1)}) = \nabla h(x^{(k)}) - \eta \nabla f(x^{(k)}).$$

Since $\nabla h(x) = x$, this becomes

$$x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)}).$$

Takeaway

Gradient descent is mirror descent with the Euclidean mirror map

$$h(x) = \frac{1}{2} \|x\|_2^2.$$

CE11.1(b): Entropy Geometry

Task

Let

$$h(x) = \sum_{i=1}^n (x_i \log x_i - x_i), \quad x_i > 0.$$

Compute $\nabla h(x)$ and $(\nabla h)^{-1}(y)$.

Coordinate-wise derivative

For each coordinate,

$$\frac{\partial}{\partial x_i} (x_i \log x_i - x_i) = \log x_i + 1 - 1 = \log x_i.$$

CE11.1(b): Gradient and Inverse Gradient

Gradient

$$\nabla h(x) = \begin{pmatrix} \log x_1 \\ \vdots \\ \log x_n \end{pmatrix}.$$

Inverse gradient map

If $y_i = \log x_i$, then $x_i = e^{y_i}$. Hence

$$(\nabla h)^{-1}(y) = \begin{pmatrix} e^{y_1} \\ \vdots \\ e^{y_n} \end{pmatrix}.$$

Notation warning

Write $(\nabla h)^{-1}$, not ∇h^{-1} . We invert the gradient map.

CE11.1(c): Multiplicative Update

Task

Use the entropy mirror map to derive the coordinate-wise mirror descent update.

Start from the dual update

For coordinate i ,

$$\nabla h(x^{(k+1)})_i = \nabla h(x^{(k)})_i - \eta \nabla f(x^{(k)})_i.$$

Since $\nabla h(x)_i = \log x_i$,

$$\log x_i^{(k+1)} = \log x_i^{(k)} - \eta \nabla f(x^{(k)})_i.$$

CE11.1(c): From Logs to Multiplicative Weights

Exponentiate both sides:

$$x_i^{(k+1)} = \exp \left(\log x_i^{(k)} - \eta \nabla f(x^{(k)})_i \right).$$

Therefore,

$$x_i^{(k+1)} = x_i^{(k)} \exp(-\eta \nabla f(x^{(k)})_i).$$

Connection to Hedge

If $\nabla f(x^{(k)})_i$ is interpreted as the loss of action i , the update multiplies each weight by $e^{-\eta \text{loss}_i}$.

Exercise CE11.2: Projected Gradient Descent

Consider minimizing a convex, differentiable function $f(x)$ over the closed convex set $X = \{x \in \mathbb{R}^2 \mid \|x\|_2 \leq 1\}$.

- (a) Formulate the update rule for Projected Gradient Descent using the projection operator $\Pi_X(\cdot)$.
- (b) Give an explicit formula for the Euclidean projection operator $\Pi_X(y)$ for any point $y \in \mathbb{R}^2$.
- (c) Let $f(x) = (x_1 - 2)^2 + (x_2 - 2)^2$. Perform one step starting at $x^{(0)} = (0, 0)^T$ with $\eta = 0.5$.

CE11.2: Problem Formulation

Optimization over a constrained set

Minimize

$$f(x) = (x_1 - 2)^2 + (x_2 - 2)^2$$

over the unit disk

$$X = \{x \in \mathbb{R}^2 \mid \|x\|_2 \leq 1\}.$$

Why projection is needed

The unconstrained gradient step may leave X . Projected Gradient Descent first takes the gradient step and then maps the result back to the closest feasible point.

CE11.2(a): Projected Gradient Descent Update

Two-stage update

First take the usual gradient step:

$$y^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)}).$$

Then project the result back onto X :

$$x^{(k+1)} = \Pi_X(y^{(k+1)}).$$

Projection operator

$$\Pi_X(y) = \arg \min_{x \in X} \|x - y\|_2.$$

It returns the feasible point closest to y .

CE11.2(b): Projection onto the Unit Disk

Unit disk

$$X = \{x \in \mathbb{R}^2 \mid \|x\|_2 \leq 1\}.$$

Projection formula

$$\Pi_X(y) = \begin{cases} y, & \text{if } \|y\|_2 \leq 1, \\ \frac{y}{\|y\|_2}, & \text{if } \|y\|_2 > 1. \end{cases}$$

Why?

If y is already feasible, it is its own closest feasible point. If y is outside the disk, the nearest feasible point lies on the ray from the origin through y , exactly on the boundary.

CE11.2(c): Step-by-Step Execution

1. Compute Gradient

$$\nabla f(x) = \begin{pmatrix} 2(x_1 - 2) \\ 2(x_2 - 2) \end{pmatrix}$$

At $x^{(0)} = (0, 0)^T$: $\nabla f(0, 0) = \begin{pmatrix} -4 \\ -4 \end{pmatrix}$

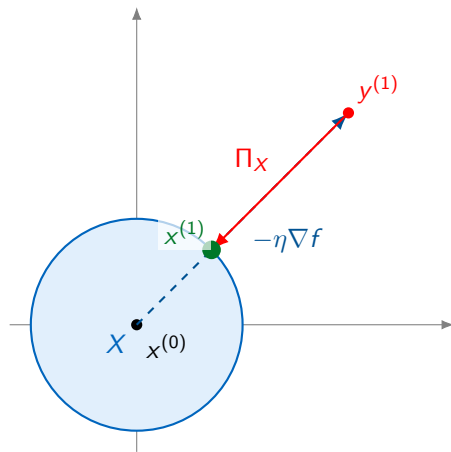
2. Unconstrained Step ($\eta = 0.5$)

$$y^{(1)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - 0.5 \begin{pmatrix} -4 \\ -4 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

Notice: $\|y^{(1)}\|_2 = \sqrt{8} > 1$. Point is outside X !

3. Projection Step

$$x^{(1)} = \Pi_X(y^{(1)}) = \frac{1}{\sqrt{8}} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$



CE11.3: Problem Statement

Exercise CE11.3: From FTL to FTRL

Consider an online linear optimization problem over the decision space $X = [-1, 1]$. In each round t , the adversary reveals a loss function $L_t(x) = g_t x$. The adversary plays the sequence of gradients: $g_1 = 0.5, g_2 = -1, g_3 = 1, g_4 = -1$.

- (a) **The Failure of FTL:** The Follow-The-Leader algorithm chooses $x_t = \arg \min_{x \in X} \sum_{i=1}^{t-1} g_i x$ (with $x_1 = 0$). Compute the sequence of decisions x_t , the cumulative loss of FTL, and the overall regret.
- (b) **The "Cheating" Baseline:** The hypothetical Be-The-Leader (BTL) strategy cheats: $x_t^{BTL} = \arg \min_{x \in X} \sum_{i=1}^t g_i x$. Compute the decisions and the regret of the BTL strategy.
- (c) **The Cure (FTRL):** To prevent the wild oscillations of FTL, we introduce Follow-The-Regularized-Leader with $h(x) = \frac{1}{2}x^2$. The update rule is:

$$x_{t+1} = \arg \min_{x \in X} \left(\sum_{s=1}^t g_s x + \frac{1}{2\eta} x^2 \right)$$

Derive the closed-form expression for x_{t+1} and discuss the connection to OGD.

CE11.3: Online Optimization Setup

Decision and loss

At each round t , choose $x_t \in X = [-1, 1]$. Then a linear loss is revealed:

$$L_t(x_t) = g_t x_t.$$

Adversarial sequence

$$g_1 = 0.5, \quad g_2 = -1, \quad g_3 = 1, \quad g_4 = -1.$$

Regret

Compare the algorithm to the best fixed decision in hindsight:

$$\text{Regret}_T = \sum_{t=1}^T g_t x_t - \min_{x \in [-1, 1]} \sum_{t=1}^T g_t x.$$

CE11.3(a): Follow-The-Leader Rule

FTL decision

Before seeing g_t , choose the decision that would have minimized all previous losses:

$$x_t = \arg \min_{x \in [-1,1]} \sum_{i=1}^{t-1} g_i x.$$

Tie-breaking

At $t = 1$, the past sum is zero. We use the tie-breaking convention

$$x_1 = 0.$$

CE11.3(a): Linear Minimization Over $[-1, 1]$

For a scalar G , the function $x \mapsto Gx$ is linear.

Minimizer

$$\arg \min_{x \in [-1, 1]} Gx = \begin{cases} -1, & G > 0, \\ 1, & G < 0, \\ \text{any } x \in [-1, 1], & G = 0. \end{cases}$$

Sign intuition

Positive G pushes the minimizer to the left endpoint; negative G pushes it to the right endpoint.

CE11.3(a): FTL Decisions

Round t	Past sum $\sum_{i=1}^{t-1} g_i$	FTL x_t	g_t	Loss $g_t x_t$
1	0	0	0.5	0
2	0.5	-1	-1	1
3	-0.5	1	1	1
4	0.5	-1	-1	1

Cumulative FTL loss

$$\sum_{t=1}^4 g_t x_t = 0 + 1 + 1 + 1 = 3.$$

CE11.3(a): FTL Regret

The total gradient is

$$\sum_{t=1}^4 g_t = 0.5 - 1 + 1 - 1 = -0.5.$$

The best fixed decision in hindsight solves

$$\min_{x \in [-1, 1]} (-0.5)x.$$

Since the coefficient is negative, the minimum is attained at

$$x^* = 1, \quad \sum_{t=1}^4 g_t x^* = (-0.5) \cdot 1 = -0.5.$$

FTL regret

$$\text{Regret}_4^{\text{FTL}} = 3 - (-0.5) = 3.5.$$

CE11.3(b): Be-The-Leader Rule

BTL decision

BTL chooses the decision that minimizes losses including the current round:

$$x_t^{\text{BTL}} = \arg \min_{x \in [-1,1]} \sum_{i=1}^t g_i x.$$

Important interpretation

BTL is not a valid online algorithm because it uses g_t before choosing x_t . It is a comparison tool that shows what happens if the player were one step ahead.

CE11.3(b): BTL Decisions

Round t	Current sum $\sum_{i=1}^t g_i$	BTL x_t^{BTL}	g_t	Loss
1	0.5	-1	0.5	-0.5
2	-0.5	1	-1	-1
3	0.5	-1	1	-1
4	-0.5	1	-1	-1

Cumulative BTL loss

$$\sum_{t=1}^4 g_t x_t^{\text{BTL}} = -0.5 - 1 - 1 - 1 = -3.5.$$

CE11.3(c): FTRL Motivation

Problem with FTL

FTL can jump abruptly between -1 and 1 because it solves a purely linear problem over $[-1, 1]$.

FTRL idea

Add a regularization term that penalizes large movements toward the boundary:

$$\frac{1}{2\eta}x^2.$$

Effect of the regularizer

The leader is no longer chosen only by past linear losses. The quadratic term anchors the decision near zero unless the accumulated gradient is large enough.

CE11.3(c): FTRL Update

Using the regularizer $h(x) = \frac{1}{2}x^2$, FTRL chooses

$$x_{t+1} = \arg \min_{x \in [-1,1]} \left\{ \sum_{s=1}^t g_s x + \frac{1}{2\eta} x^2 \right\}.$$

Let

$$G_t = \sum_{s=1}^t g_s.$$

Then the objective is

$$G_t x + \frac{1}{2\eta} x^2.$$

This is a one-dimensional strongly convex quadratic.

So it has a unique unconstrained minimizer.

CE11.3(c): Deriving the FTRL Formula

Differentiate the unconstrained objective:

$$\frac{d}{dx} \left(G_t x + \frac{1}{2\eta} x^2 \right) = G_t + \frac{1}{\eta} x.$$

Set the derivative equal to zero:

$$G_t + \frac{1}{\eta} x = 0.$$

Solve for x :

$$x = -\eta G_t = -\eta \sum_{s=1}^t g_s.$$

Because x must lie in $[-1, 1]$, we project:

$$x_{t+1} = \Pi_{[-1,1]} \left(-\eta \sum_{s=1}^t g_s \right).$$

CE11.3(c): Connection to Online Gradient Descent

Let

$$x_t = \Pi_{[-1,1]}(-\eta G_{t-1}), \quad G_t = G_{t-1} + g_t.$$

Then

$$-\eta G_t = -\eta G_{t-1} - \eta g_t.$$

So, before projection, the new point is obtained by subtracting ηg_t , exactly like an online gradient step for the linear loss $g_t x$.

Why this matters

For bounded gradients and an appropriate choice of η , the quadratic regularizer prevents the unstable boundary oscillation seen in FTL and yields sublinear regret.

Optional: CE11.3(c): FTRL Decisions ($\eta = 0.5$)

Let's trace the regularized algorithm with a learning rate of $\eta = 0.5$. The update rule implies $x_t = \Pi_{[-1,1]} \left(-0.5 \sum_{s=1}^{t-1} g_s \right)$.

Round t	Past sum $\sum_{i=1}^{t-1} g_i$	FTRL x_t	g_t	Loss $g_t x_t$
1	0	0	0.5	0
2	0.5	-0.25	-1	0.25
3	-0.5	0.25	1	0.25
4	0.5	-0.25	-1	0.25

Note: All unconstrained updates lay within $[-1, 1]$, so the projection $\Pi_{[-1,1]}$ simply returned the value itself.

Cumulative FTRL loss

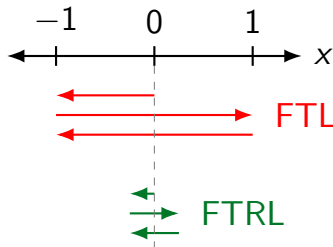
$$\sum_{t=1}^4 g_t x_t = 0 + 0.25 + 0.25 + 0.25 = 0.75.$$

Optional: CE11.3(c): FTRL Regret & Stabilization

FTRL Regret

Recall that the loss of the best fixed decision in hindsight was -0.5 .

$$\text{Regret}_4^{\text{FTRL}} = 0.75 - (-0.5) = 1.25$$

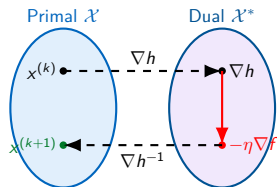


The Power of Regularization

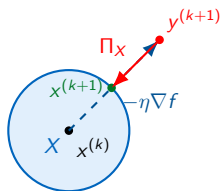
- **FTL Regret:** 3.5
- **FTRL Regret:** 1.25

By hedging its bets and choosing fractional points instead of jumping to the extreme boundaries, FTRL neutralizes the adversary's out-of-phase trap!

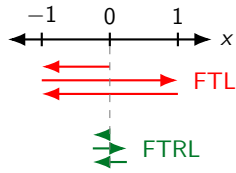
1. Mirror Descent



2. Projected Gradients



3. FTRL Stabilization



Discussion & Questions